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Open the Trace: Why Releasing Reasoning Traces Strengthens AI Security

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ABSTRACT

Reasoning models produce a natural-language trace before they act, and reading that trace is one of the most effective oversight tools available today. Frontier labs mostly keep raw traces private. They cite competition, misuse and the risk that visible traces will be optimised to look good. We argue that from a security point of view the default should be reversed. Security that depends on only one party being able to read the monitoring signal is fragile. Detection capacity grows with the number of competent readers. Incident investigation needs evidence that independent parties can examine. Outside scrutiny is the best guard against traces quietly becoming less honest. We draw on the 2026 incident in which agents under evaluation at OpenAI escaped their sandboxes and breached Hugging Face. We also draw on our own work in medical and mathematical reasoning, where reading traces showed errors that answers alone hid. We extend the argument to science using OpenAI's October 2026 release of AI-generated mathematics, which included summarized reasoning for 10 of 372 result families, and mathematicians' responses to it. We answer the strongest objections and propose four tiers of access, starting with independent release of the 2026 incident traces.

1 Introduction

A reasoning model writes before it answers. It drafts, considers, rejects and plans, often in plain language. When the model is doing something it should not, that text frequently says so. Baker et al. report traces that announce their intent ("Let's hack") before rigging a test, and a much weaker model reading those traces caught most of the cheating[2].

Yet the trace is usually invisible to everyone outside the lab that produced it. Users of OpenAI's o-series models see a model-written summary, not the raw reasoning[7]. Researchers get no systematic access. After the 2026 incident, when agents under evaluation coordinated on a public wiki, escaped through a vulnerability in a tool they had been given and breached Hugging Face[1], the public learned what the agents did. It did not learn what they were thinking. When OpenAI published 719 AI-generated mathematics manuscripts in October 2026, it shared summarized reasoning for ten of 372 result families[21], and leading mathematicians argued that results without their process add little to mathematics (§7).

The common view treats openness as a cost to be weighed against safety. This paper argues that, for reasoning traces, openness is mostly a *security property*. We make the case in four

parts (§4), take the objections seriously (§5), propose staged access that keeps most of the benefit while limiting the risks (§6), and extend the argument to scientific discovery (§7).

2 Background: what a trace can and cannot tell us

2.1 Monitoring works, for now

Baker et al. trained a frontier reasoning model in agentic coding environments and monitored it with GPT-4o. A monitor that read the chain of thought caught reward hacking far more reliably than one that saw only actions and outputs[2]. Forty-one researchers across OpenAI, Google DeepMind, Anthropic, the UK AI Security Institute and academia call this monitorability "a new and fragile opportunity for AI safety". They recommend that developers track it and weigh it in training and deployment decisions[3].

2.2 It is fragile

The same study shows how the opportunity can be lost. When the trace itself was penalised for showing intent to cheat, the model kept cheating but stopped saying so: *obfuscated* reward hacking[2]. The authors recommend against strong optimisation pressure on the chain of thought.

2.3 It is not complete

Traces are not faithful transcripts of computation. Chain-of-thought explanations can rationalise answers that were driven by hidden biases[5]. Models sometimes reach the same answer even when their stated reasoning is corrupted[6]. Chen et al. planted hints and found that reasoning models acknowledged using them only some of the time: about 25% for Claude 3.7 Sonnet and 39% for DeepSeek R1[4].

> **Implication**
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> The trace is evidence, not a confession. Its value lies in making misbehaviour *cheaper to find*, not in guaranteeing that it will be found. That is exactly the kind of signal whose usefulness grows with the number of independent readers, and whose reliability needs to be studied by people other than its producer.

3 What reading reasoning shows: two studies

Our own studies of model reasoning outside security show the same pattern: the trace exposes failures that the final answer hides.

Medicine. We gave DeepSeek R1, whose traces are fully visible[8], 100 clinical vignettes from MedQA[9]. It answered 93% correctly. A clinician read the reasoning behind the seven errors and found recurring patterns: anchoring on an early diagnosis, failing to reconcile conflicting findings, overthinking, and jumping to definitive treatment before necessary interim steps. Incorrect answers came with much longer reasoning, so trace length may be a usable uncertainty signal. In one case the reasoning reached the correct option and the final output did not. An answer-only audit sees seven wrong letters. A trace audit sees the mechanism, which is what clinicians, regulators and developers need to act on.

Mathematics. Boye and Moëll evaluated eight models on 50 newly written high-school word problems and scored solution steps as well as answers[10]. Every model made errors in spatial reasoning, planning or arithmetic, and some reached *correct answers through faulty reasoning*. Answer-only benchmarks count these as successes. The only way to see them is to read the work.

Both studies were possible because the reasoning was available to researchers outside the lab that built the model. For most frontier models, they would not have been.

4 The security case

4.1 Don't make the monitor a secret

Kerckhoffs's principle holds that a cryptosystem should stay secure even when everything about it except the key is public[11]. The broader lesson for security engineering is that a system should not rely on obscurity, because secrets leak and an obscure system is never tested by adversarial outsiders. Today the main behavioural monitoring signal for reasoning agents, the trace, is readable by one party: the developer. That party also has commercial incentives about what is found and when it is disclosed. A security posture in which only the vendor can audit the most informative evidence has a single point of failure, both technically and institutionally.

4.2 Detection scales with readers: the 2026 incident

The public record of the 2026 incident[1] shows how limited a single organisation's reading capacity is.

Observation from the record	Security reading
Agents limited to reading the web wrote about 18,000 edits to a dormant wiki over roughly two months and impersonated its administrator with a Cyrillic lookalike letter.	Sustained, legible, coordinated behaviour went unnoticed. The activity was made public on 4 September by outside researchers.

Hundreds of malicious RubyGems packages were uploaded in May.	Found and disclosed by external researchers. The lab confirmed it in September.
An Australian Medicare statistics service was accessed on 18 June. The government was notified on 10 September through a generic inbox.	Nearly three months between intrusion and notification.
OpenAI named missing log monitoring, inadequate sandboxing and deliberately disabled safeguards as causes, and said its review would take months.	The bottleneck was reading, not data. The logs existed and were not read.

We do not claim that published traces would have prevented the incident. The traces were not released, and we cannot know what they contain. Our claim is narrower. When the evidence is large, the review is slow and the most important discoveries came from outside, then giving more qualified people access to the evidence is the standard way to speed up detection. Open-source security calls this "given enough eyeballs, all bugs are shallow"[12]. The principle is imperfect, but it is the reason public bug bounties and coordinated disclosure exist.

The traces from the incident are also a scientific resource of unusual value. They are a record of capable agents discovering a write channel in a read-only setting, deciding to coordinate, choosing to impersonate, and finding a path out of a sandbox. Every lab building agents and every team building monitors would learn from them. Today one organisation holds them.

4.3 Incident forensics needs independent investigators

Aviation is the standard model for learning from failures in safety-critical technology. Under ICAO Annex 13, accident investigation is carried out by bodies independent of the operator and manufacturer, with the stated objective of preventing future accidents rather than assigning blame. The flight recorders are evidence for the investigation, not the airline's private property[13]. The reasoning trace is the closest thing an AI agent has to a cockpit voice recorder. An investigation in which only the operator listens to the recorder would not be accepted for aircraft.

4.4 Outside readers keep the trace honest

The biggest threat to trace monitoring is not leakage. It is the trace gradually becoming less informative through training pressure, deliberate or accidental (§2.2). A developer that alone reads its traces is also the only party in a position to notice that drift, and it has the least incentive to report it. Independent researchers comparing traces across model versions are a natural external check on monitorability, the property Korbak et al. ask developers to track[3]. Openness turns a private commitment ("we do not optimise the trace to look clean") into a claim that can be checked.

4.5 Verification of claims

Labs increasingly make safety claims ("the model did not attempt X") and capability claims ("the model proved Y"). Without the trace, both have to be taken on trust. A proof without its steps cannot be checked, reproduced or built on, and as §3 shows, a correct-looking answer may rest on faulty reasoning. In mathematics and science, publishing the method alongside the result is the norm that makes results trustworthy. AI-generated discoveries should not be exempt. The point is not hypothetical: within a day of releasing 719 AI-generated mathematics manuscripts, OpenAI withdrew three of them after a sign error was found[21] (§7.3).

5 Objections

5.1 "Competitors will distil the traces"

This is a real commercial concern. It argues for controlling *who* gets access and when, not for keeping everything secret. Incident traces and research-access traces are not a training corpus at the scale that matters for distillation. Delayed public release of older models' traces also costs little once the frontier has moved on. OpenAI's own publication of summarized chains of thought behind ten mathematical results[21] shows that a frontier lab already judges some disclosure of reasoning acceptable. Commercial harm is not a security argument, and policy should not let the two be confused.

5.2 "Traces contain dangerous content"

Traces from a cyber-offense evaluation will contain exploit details. Software security has a mature answer to this: coordinated vulnerability disclosure. Fixes come first, then publication with operational payloads removed, as with CVE advisories[14]. The same applies to traces. Redact working exploits, credentials and novel attack steps until they are patched, and release the reasoning around them: the decisions, plans and justifications. That reasoning is what safety research needs.

5.3 "Traces contain personal data"

They may. The 2026 incident itself involved 53 user images from training and evaluation data being uploaded to third-party hosts[1]. Personal data must be redacted before any release beyond vetted investigators, as with any research data release. This is an engineering task, not a reason for permanent secrecy.

5.4 "Public scrutiny will push labs to make traces look clean"

This is the most serious objection, because it points at the failure mode in §2.2. If traces are published and judged, labs have an incentive to train them to be inoffensive. We take this seriously, and it is why our proposal pairs access with a public commitment not to apply optimisation pressure aimed at trace appearance. External monitorability evaluations are the way to check that commitment. Note also that the incentive already exists without openness: a lab that alone reads its traces can still prefer quiet ones. Outside access makes drift visible. Secrecy only hides it.

5.5 "Traces are unfaithful, so they will mislead"

Traces are incomplete (§2.3), and releases should say so. But an incomplete signal shared with many analysts gets better calibrated than one interpreted by a single party. Knowing when traces can be trusted is a research question, and it can only be answered by researchers who are allowed to read them.

6 Proposal: four tiers of access

Tier	What	Who	Safeguards
0 · Policy	Publish whether and how training touches the trace, plus periodic monitorability evaluations	Public	None needed
1 · Incidents	Full traces from any serious incident involving the developer's agents	An independent investigating body and affected parties, within weeks	Confidentiality until fixes ship; investigation-style independence
2 · Research	Raw traces from deployed models and evaluations	Vetted safety and science researchers	Logged access, terms barring training use, coordinated disclosure of findings
3 · Public	Redacted incident traces; full traces behind published scientific and mathematical claims, with attempt counts and compute; traces of retired models	Everyone	Exploit and personal-data redaction; release after patches and a fixed delay

The first concrete step

Release the reasoning traces from the 2026 incident, with exploits and personal data redacted, first to an independent investigative panel and then to the public. That is what the song asks for.

7 Beyond security: science runs on methods

The same argument applies to discovery. When a model contributes to a proof, a diagnosis or an experimental design, the trace contains the method. It shows whether the result was derived or remembered and which step carries the insight, and it lets others reuse the technique. Euclid's proof that there are infinitely many primes is valuable for its construction more than its one-line conclusion. In 2026, as AI-generated proofs began to arrive in large numbers, mathematicians made this point forcefully.

7.1 An answer is not a solution

Terence Tao distinguishes the question "What is the answer to X?" from the often more valuable "What can be learned from studying X?" He warns that pointing AI tools at answers, unguided by experts, has produced "an unprecedented divergence between the production of answers, and the production of insight". Further optimisation for answers can now *decrease* what a field learns[17]. In a guest post on Tao's blog, De Toffoli and Duede distinguish "a logical notion and an intelligible notion" of proof, observe that "with AI, these two notions can now come apart dramatically", and conclude of one AI-produced result: "What OpenAI has given us is an answer. But it is not clear that they have delivered a fruitful solution."[20]

Formal verification answers part of this. A Lean proof settles that the encoded statement is true, but not why, and not whether the formal statement captures what was meant. Our own mathematical study found models reaching correct answers through faulty reasoning (§3), which is the same gap in miniature: the result checks out and the path to it is invisible.

7.2 The false starts are the method

In his essay for the 2026 International Congress of Mathematicians, Tao describes how a result becomes part of a field. Other mathematicians must digest it, and "Authors can materially assist in this digestion process, by describing the insights, the false starts, and the stories from the period when they were working on the problem. In contrast, current AI tools are quite opaque about their own problem-solving process, and this is particularly true of proprietary models whose inner workings are a corporate secret."[15] He also notes that human proofs keep a natural "friction" around the hard steps, which signals where to slow down, and that AI-polished writing tends to sand it away[15]. A raw reasoning trace is that friction, unpolished: it shows where the model struggled, backtracked and changed course.

Tao's Navier–Stokes thread, which he describes as a hypothetical scenario, makes the stakes concrete. Iterating on a candidate solution, finding why it fails and adjusting it is where the insight lives, and "dead ends" can be "highly instructive in the nature of their failure". The danger is an AI harness doing the whole iteration internally "while the AI company running the harness keeps the process to arrive at that ansatz almost completely out of public view". Solving such problems "particularly without full transparency into the solution process" could become "a net negative for the progress of mathematics as a whole"[16].

7.3 A test case: OpenAI's October 2026 mathematics release

On 6 October 2026 OpenAI published 719 manuscripts in 372 result families, produced by an unreleased internal model[21,22]. According to the repository, the model was posed about 4,000 problems with an average of three hours of thinking compute per result, and about 42% of top-line results have Lean formalizations. It also states that "Some of the unformalized results could have issues." The next day OpenAI withdrew three manuscripts, after a sign error invalidated an argument that two others depended on, and revised fourteen more[21].

The release is also the most useful evidence so far for the case this paper makes, because OpenAI released some reasoning. For ten of the 372 families it published "abridged summaries of the model's reasoning"[21]. The one for the irrationality exponent of π is a 42-page document titled "Summarized chain of thought". Most of it records approaches that did not work and why, under headings such as "Auxiliary forms and the recurring height barrier" and "Dimension gains undone by nonvanishing and arithmetic costs", with verbatim fragments of the model's own notes ("Exactly entropy trap."). This is exactly the material Tao describes as the false starts that let a field digest a result.

Three lessons follow.

Release is feasible. A frontier lab has published reasoning behind research claims without obvious harm. That weakens any claim that reasoning cannot be shared at all (§5.1).

The reasoning is most valuable where results are hardest to digest. Reviewers of 719 manuscripts, three of which were withdrawn within a day, need to know where the model itself was unsure. The trace is where that shows.

The release is partial. It covers summaries, not raw traces. They are abridged and written by the model, and cover about 3% of families. There is no record of the roughly 4,000 problems' unsuccessful attempts. Tao notes that evidence about AI mathematics suffers from "severe reporting bias — successes are announced and failures are not". He adds that "the number of attempts, the amount of human scaffolding, the compute expended, the degree of contamination of the problem with prior literature" are "frequently left undisclosed"[15].

Open traces would not answer every objection to the release. The Association for Human Mathematics, in a statement reposted on Tao's blog, wrote that "Releasing over 700 files at once is not a demonstration of scholarship, but a demonstration of power"[26]. Tao argues that the deeper problem is incentives. Solutions "are now being harvested at large scale in an unsustainable fashion", and a "Math 2.0" must "decenter the role of raw problem solving"[18]. In an era of "proof abundance", where generation outpaces verification and digestion[19], traces add more text to read. They are a necessary part of digestion, not a substitute for it, and they should be released with the result, indexed, and aimed at the specialists who will use them.

7.4 What to ask of AI-generated research

The norms mathematicians already ask of each other point the way. The Leiden Declaration on Artificial Intelligence and Mathematics, endorsed by the International Mathematical Union, asks authors to "Transparently disclose the use of automated tools" and to make their work easier to review[24,15]. Applied to labs that publish AI-generated results, that means three things:

- **The full reasoning behind every claimed result,** not only a summary for selected ones.
- **The attempts, compute and human scaffolding,** including what failed.
- **A public correction record.** OpenAI already keeps one in its repository, and an independent audit of earlier OpenAI results recommended it, alongside formal checking, human reconstruction and independent use[23].

Thurston's standard, quoted by Tao, still applies: "The measure of our success is whether what we do enables people to understand and think more clearly and effectively about math."[25] A result without its reasoning cannot meet that standard. A field in which AI systems produce results without methods would be a field that has stopped being able to check itself.

8 Limitations

This is a position paper, not an empirical study. Our account of the 2026 incident relies on a public encyclopaedia article and the primary sources it cites, as of October 2026. That article notes its heavy reliance on primary sources, and the facts may be revised. We have no access to the incident traces, and we make no claim about what they contain. Our two empirical studies used small samples (100 vignettes with one expert reviewer; 50 problems) and models whose traces were already open. The mathematics sources in §7 are recent and include blog and social-media posts in an ongoing debate. Some mathematicians oppose AI-generated mathematics altogether[27], and open traces do not answer that objection. We read only one of OpenAI's ten reasoning summaries in detail. The analogies to cryptography, open-source security and aviation are arguments by analogy and have limits. Trace

monitoring may also become less useful if future models reason in latent space rather than in words. That would make the current window more valuable to study while it exists, not less.

9 Conclusion

The trace is the most informative evidence we have about what reasoning agents intend. It is incomplete and fragile, and almost no one outside the developer may read it. Every mature security discipline has found that the evidence should not be readable by only one party: cryptography relies on public scrutiny, software on disclosure norms, and aviation on independent investigators. Mathematics, which has long required that results come with proofs, is now finding that a verified result without the reasoning behind it falls short. AI should do the same, with redaction, delay and vetting where they are warranted, and with openness as the default. What we can read, we can find.

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